# Inverting a Foundation Brain Encoder: Manuscript

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## Predictable Is Not Recoverable: Conditioning and Non-Convexity of Inverting a Foundation Brain Encoding Model

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**Abstract.** Deep brain encoding models now predict whole-cortex fMRI responses to natural stimuli, and decoding is often assumed to follow from encoding accuracy. We test that assumption by inverting TRIBE v2, a 177 M-parameter open-weight trimodal encoder, by gradient descent on its 1 152-dimensional multimodal representation, re-entered at the differentiable boundary after feature aggregation. With exact synthetic ground truth, encoding error falls 28.4-fold across optimisation budgets while recovery saturates (cosine 0.944); the final halving of error buys +0.015. Fit-matched solutions from different initialisations disagree (pairwise cosine down to 0.835; 0.817 ± 0.007 across four ground truths). Over four ground truths, the error along the straight line between two distinct, equally good blind solutions rises 2.4–8.2× (median 4.8×), and blind starts settle in a minimum with 4.2× the error of one reached from near the truth, separated from it by an 8.1 ± 3.2× barrier: the minima are distinct, not a flat null space. The solution difference lies along directions less sensitive than every one of 200 random directions (2.15 ± 0.17× below the median). Gain is identifiable, conventional priors degrade recovery, and gradient inversion exceeds a ridge decoder about fifteen-fold. Across six anatomical networks, the most responsive (visual, 1.60× the cortical mean) recovers worst (−0.125 against size-matched random vertices) and the least responsive (somatomotor) recovers best (+0.102). On real fMRI from a held-out story (LeBel et al., 2023), the encoder reaches r = 0.48 on its best-predicted vertices (56% of the noise ceiling), and inverting measured responses identifies the heard 100-s segment in 4 of 5 cases among 45 (chance 2.2%), while the recovered representation has cosine 0.02 to the truth; even inverting the model's own noise-free prediction recovers only 0.22. Encoding accuracy does not certify decoding: the inverse identifies stimuli but does not reconstruct their representation.

**Keywords:** brain encoding model, inverse problem, fMRI decoding, identifiability, conditioning, foundation model, TRIBE v2.

## I. Introduction

A brain encoding model maps a stimulus to the brain response it evokes; decoding runs the other way. The encoding route to decoding has framed the field for fifteen years [1]: fit a forward model, then identify or reconstruct stimuli by asking which input best explains a measured response [7], [4]. Encoders have since grown from linear maps on hand-built features [8], [2] to deep networks trained on hundreds of hours of fMRI, and their accuracy keeps rising with data and model size [3], [5].

Forward accuracy, however, says nothing about the inverse. A problem is well posed only if its solution exists, is unique and depends stably on the data [13]; a forward map can be near-perfect and still violate the last two [12]. For linear inverted encoding models this was shown analytically: they recover the stimulus only up to an invertible linear transform [9]. No study has asked the same questions of a deep, nonlinear foundation encoder, where no closed-form answer exists.

TRIBE v2 [5] makes the question answerable. It is an open-weight trimodal transformer (177 M parameters) that predicts responses at 20 484 cortical vertices from video, audio and language features, and it is differentiable once its feature extractors have run. We treat it as a forward operator and characterise its inverse with exact ground truth, then test the conclusions on real fMRI.

Our contributions:

1. **The first uniqueness and conditioning analysis of a deep foundation brain encoder.** A literature search on 9 October 2026 (Section II) found no published inverse or identifiability analysis of such a model; the two TRIBE v2 follow-ups [10], [11] use it for feature visualisation and synthetic training data.
2. **A differentiable re-entry point** into the TRIBE v2 pipeline after feature aggregation, through which gradients reach every entry of the representation.
3. **Encoding accuracy and decodability dissociate:** encoding error falls 28.4-fold while recovery saturates.
4. **Incomplete recovery reflects non-convexity, not a null space,** and the residual error concentrates in the operator's least-sensitive directions.
5. **Predictable is not recoverable, region by region:** the most responsive network is the hardest to decode from, and the least responsive the easiest.
6. **Real-fMRI confirmation:** inverting measured responses identifies which story segment was heard while recovering little of its representation, the pattern the synthetic analysis predicts.

## II. Related Work

**Encoding and decoding.** Voxelwise encoding models predict fMRI from stimulus features, from word co-occurrence [8] to semantic maps of natural speech [2] and language-model features whose accuracy scales log-linearly with model size [3]. Identification compares a measured response with responses predicted for candidate stimuli [7]; reconstruction searches stimulus space for the input whose predicted response fits best, as in continuous language decoding guided by a language model [4]. Non-invasive decoders trained end to end reach useful accuracy with MEG but much less with EEG [15], [16], and shared-subject models reduce the data each subject needs [17]. In every case the decoder is evaluated by proxies (rank, retrieval, word error), because the true representation behind real fMRI is unknown.

**Inverting encoders.** Inverted encoding models reconstruct channel responses from voxel patterns [22]; Gardner and Liu proved that such linear models recover the stimulus only up to an invertible linear transform, so a reconstruction reflects the model as much as the brain [9]. In computer vision, inverting deep networks from their activations exposes what information each layer keeps [29]. Neither line of work addresses a deep, nonlinear brain encoder.

**TRIBE v2.** TRIBE won the Algonauts 2025 challenge [6]; TRIBE v2 extends it into an open-weight foundation encoder trained on 451.6 h of fMRI and evaluated on 1 117.7 h from 720 subjects [5]. Its authors frame it for in-silico neuroscience. Two follow-ups appeared in 2026: Bladon and Bent optimise inputs to maximise a region's predicted response and recover known cortical selectivity [10]; Benchetrit et al. use TRIBE v2 forwards to synthesise fMRI that augments the training of brain-to-image decoders [11]. Neither asks whether the input can be recovered from a response, or whether the answer is unique.

**Novelty search.** On 9 October 2026 we searched arXiv, Google Scholar and the general web for "TRIBE v2" and for invert, inversion, identifiability, conditioning and non-convex combined with "brain encoding model" or "fMRI encoder", including any work that inverts, optimises the input of, or analyses the identifiability of a brain encoding model. Table I places the closest work against the four properties this study combines.

**Table I.** Closest prior work (● yes, ○ no).

| Study | Nonlinear deep encoder | Optimises encoder input | Tests uniqueness / conditioning | Region-wise recoverability |
|---|---|---|---|---|
| Kay et al. 2008 [7] | ○ | ○ | ○ | ○ |
| Gardner & Liu 2019 [9] | ○ | ● | ● | ○ |
| Tang et al. 2023 [4] | ○ | ● | ○ | ● |
| d'Ascoli et al. 2026, TRIBE v2 [5] | ● | ○ | ○ | ○ |
| Bladon & Bent 2026 [10] | ● | ● | ○ | ○ |
| Benchetrit et al. 2026 [11] | ● | ○ | ○ | ○ |
| **This work** | ● | ● | ● | ● |

Tang et al. score word sequences with a linear encoder on GPT features and compare cortical networks, so they meet two columns; their encoder is linear in its features, and uniqueness is not tested.

## III. Methods

### A. Forward operator

TRIBE v2 [5] maps a multimodal representation $x \in \mathbb{R}^{T \times H}$, $H = 1\,152$ (text, audio and video slots of 384 each, at 2 Hz), to a predicted response $y \in \mathbb{R}^{V \times T'}$ on the fsaverage5 surface [24], $V = 20\,484$ vertices, $T' = 100$ output steps at 1 Hz. The network (177.21 M parameters) is a transformer encoder (127.47 M), a low-rank head ($1\,152 \to 2\,048$), a subject-layer predictor (41.97 M; the public release ships only the shared average-subject layer) and a temporal pooler. All weights are frozen.

**Training data of the encoder.** TRIBE v2 was trained on four public 3T datasets: 25 healthy adults, 2 618 sessions, 451.6 h of fMRI, recorded during passive viewing or listening with no task [5].

CNeuroMod/Algonauts 2025 [30]: 4 subjects watched *Friends* (seasons 1–6) and four feature films (1 408 sessions, 268.7 h, TR 1.49 s). BOLD Moments [31]: 10 subjects watched 1 102 three-second video clips (520 sessions, 61.9 h, TR 1.75 s). LeBel et al. [14]: 8 subjects listened to spoken autobiographical stories (432 sessions, 85.8 h, TR 2 s). Wen et al. [32]: 3 subjects watched natural movie clips (258 sessions, 35.2 h, TR 2 s). It was evaluated without retraining on NNDb, Le Petit Prince, Narratives and HCP 7T (695 further subjects, 666.1 h), 720 subjects and 1 117.7 h in all [5]. Our real-data subject (UTS03) is a LeBel training subject, but the story we invert, *wheretheressmoke*, is TRIBE v2's own held-out test split, so the encoder never saw those responses.

**Why fMRI rather than EEG.** fMRI resolves millimetre-scale cortical patterns at 20 484 vertices [33], whereas 32–128 scalp electrodes record a volume-conducted mixture; decoding through EEG would add a second ill-posed inverse, source localisation [34], in front of the one studied here. No open EEG encoder of naturalistic stimuli exists at TRIBE v2's scale; the same decoder reaches a 32% character error rate from MEG but 67% from EEG [16]; and the open simultaneous EEG–fMRI dataset has no sustained speech [35]. EEG's millisecond resolution and portability matter for a deployed device, not for measuring how invertible an encoder is; the method transfers to EEG once a comparable encoder exists.

## B. The inversion boundary

The public inference path is not differentiable end to end: stimuli pass through speech synthesis, transcription and frozen feature extractors, and prediction runs under inference mode. The network's forward pass consumes its input batch at one point, the aggregation of extractor features into x. Everything after it is tensor arithmetic, so we take x as the free variable and re-enter the network there (Fig. 1):

```
f(x) = \mathrm{pool}\big(\mathrm{predictor}(\mathrm{head}(\mathrm{encoder}(x)))\big)
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Gradients reach every entry of x ($\|\partial y/\partial x\| = 7.97 \times 10^3$, no zero entries). Re-entering at this boundary reproduces the released model's predictions exactly (Pearson r = 1.0000 on real stimuli). Results therefore characterise the operator from the aggregated representation onward.

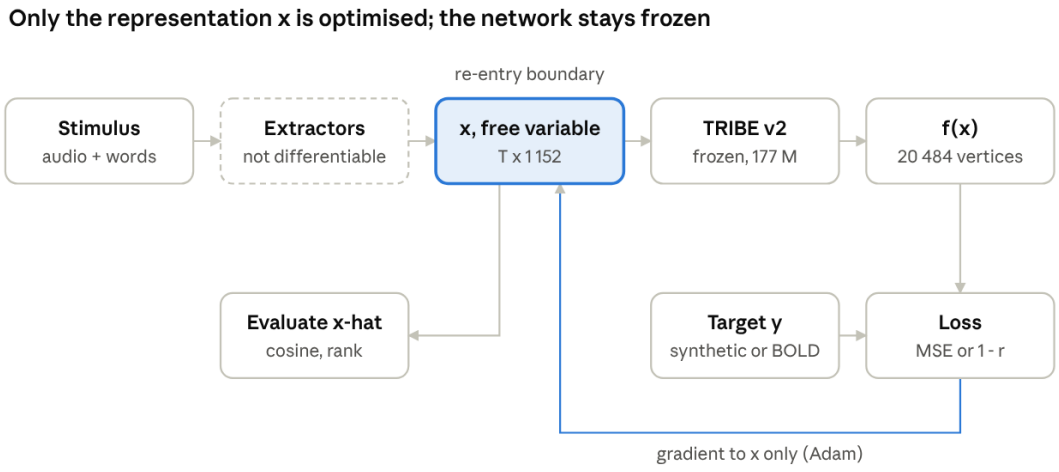



Fig. 1 · inversion pipeline: re-entry after feature aggregation, frozen network, gradient to x only

### C. Inversion

Given a response y, the representation is recovered by

$$\hat{x} = \arg\min_{x} \; \lVert f(x) - y \rVert_2^2 + \lambda R(x)$$

with Adam [18] (learning rate $5 \times 10^{-2}$, cosine annealing over the step budget), initialisation $0.01 \cdot N(0, I)$ unless stated, and R an $\ell_2$ or first-difference temporal-smoothness penalty where tested ($\lambda = 0$ otherwise). On real fMRI, where scale and offset of the measured response are arbitrary, the loss is one minus the per-vertex Pearson correlation over time, averaged over observed vertices.

### D. Synthetic protocol and metrics

Ground-truth trajectories (T = 32) are drawn in a random 8-dimensional subspace of $\mathbb{R}^H$ of TRIBE v2's own input space: eight random unit directions, each with its own cosine time course (2, 4, …, 16 cycles per 16-s window), scaled to unit standard deviation, with four seeds; i.i.d. Gaussian noise serves as an off-manifold control. This choice is representative: real story representations from the same pipeline (LeBel corpus, audio slot, 246 windows) have an effective dimensionality of 4.9 ± 2.1 with 84% of their variance in eight directions and a standard deviation of 1.10, and repeating the tests with a real 16-s representation as ground truth gave the same partial, start-dependent recovery with barriers between solutions (Section IV-C). The target is y = f(x), so recovery error is known exactly. Recovery is the mean per-time-step cosine between $\hat{x}$ and x; we also report relative error $\|\hat{x} - x\|/\|x\|$, encoding error $\|f(\hat{x}) - f(x)\|/\|f(x)\|$ and retrieval rank among other time steps. Directional sensitivity along a unit vector v is a finite-difference probe of the Jacobian:

$$s(v) = \frac{\lVert f(x + \varepsilon v) - f(x) \rVert}{\varepsilon}$$

### E. Discriminating non-convexity from non-identifiability

Inversions are started from eight initialisation distributions differing in scale and structure; two of them (the truth plus noise) serve as controls for whether a deeper minimum exists, since a real decoder never knows the truth. Only blind solutions within 2× of the best blind fit are compared, so under-converged runs cannot pose as alternative solutions; agreement is replicated over four ground truths. For two fit-matched solutions, a null space predicts that the prediction error stays flat along the straight line between them; distinct minima predict a rise in between. This test and the sensitivity probe use forward passes only and do not depend on the optimiser.

### F. Regional analysis

Six disjoint networks are defined on fsaverage5 from the Destrieux atlas [19]: left language [20] (1 282 vertices), its right homologue (1 368), auditory (277), default mode [21] (1 224), visual (1 582) and somatomotor (1 487) as a low-semantic control. For each network we measure (i) forward sensitivity per input modality (16 random directions per slot, ε = 1% of $\|x\|$), against 2 000 random vertex sets of equal size; (ii) recovery when only that network is observed (8 000 steps), against size-matched random vertex sets with a second random seed as the control spread; and (iii) the participation ratio [23] of the network's responses to 128 random input directions, a measure of how many independent directions it distinguishes.

### G. Real-fMRI validation

**Data.** Subject UTS03 of the LeBel et al. corpus [14] (OpenNeuro ds003020), six stories (66.6 min, 10 363 words, TR = 2 s). The held-out story, *wheretheressmoke*, is the mean of ten repeats and is not in TRIBE v2's training set; the other five serve as training stories for every tuning choice.

**Surface projection.** The authors' preprocessed responses (95 556 cortical voxels) are mapped to fsaverage5 with TRIBE v2's projector settings (3-mm ball around each mid-cortical vertex, 20 sample points, trilinear weights inside the cortical mask). Vertices are carried from MNI space to the subject's T1 by an ANTs SyN registration [25] and into functional voxels by the pycortex transform [26]; the series are detrended and z-scored. Validation: homotopic left–right correlation 0.635 against −0.017 for heterotopic pairs (above 97% of alternative pairings); a scrambled-vertex control falls to chance.

**Features and alignment.** Stimulus features come from TRIBE v2's own pipeline with word timings from the dataset's TextGrids: w2v-BERT 2.0 for audio [27] and Llama-3.2-3B for text [28], grouped into 100-s segments (45 across six stories). On the 15-GB GPU used here, Llama-3.2-3B was run in half precision; each story was processed in a separate run to fit memory. The offset between model time and BOLD rows is the lag that maximises mean prediction–response correlation on the five training stories; the 2 000 analysis vertices are those best predicted on the training stories.

**Evaluation.** (i) Encoding: per-vertex Pearson r on the held-out story, against a split-half noise ceiling from the ten repeats (Spearman–Brown corrected). (ii) Forward identification: each 20-s window of held-out BOLD is matched to model predictions for all 194 windows of the six stories. (iii) Inversion: each held-out 100-s segment is inverted from measured BOLD on the 2 000 vertices, optimising only the slots the features fill; $\hat{X}$ is then ranked against the true representations of all 45 segments by mean per-time-step cosine (chance 1/45). An oracle inverts the model's own prediction under the same budget, separating the operator's conditioning from measurement noise.

## IV. Results

### A. Encoding accuracy does not certify decoding

Across independent inversions at budgets from 250 to 32 000 steps, encoding error fell 28.4-fold ($7.55 \times 10^{-2}$ to $2.66 \times 10^{-3}$) while recovery cosine rose from 0.051 and saturated at 0.944. The final halving of encoding error bought +0.015 in recovery (Fig. 2). A single 64 000-step run reached 0.910, consistent with the plateau.

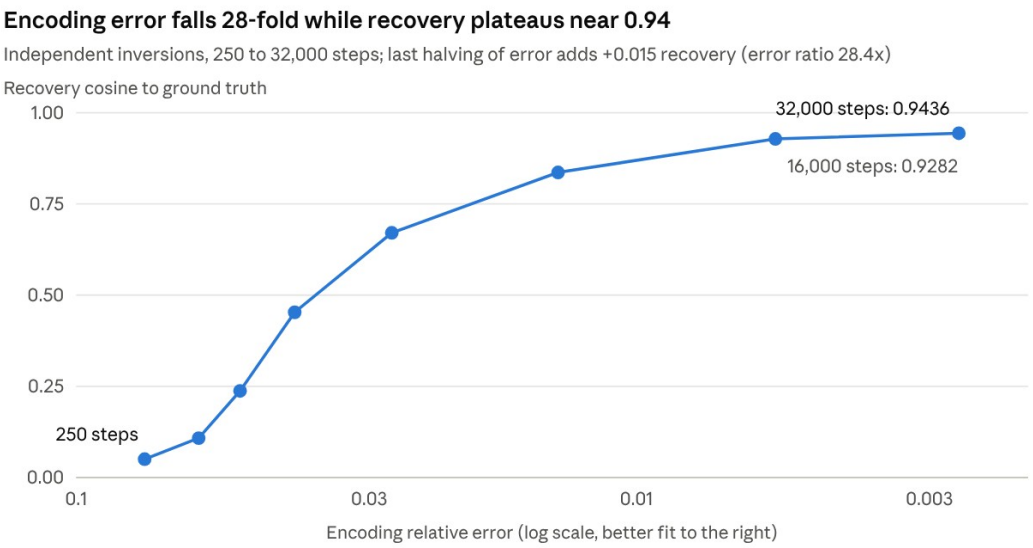



Fig. 2 · synthetic inversions of TRIBE v2, eight independent step budgets

## B. Fit-matched solutions disagree

At 16 000 steps, seven of eight initialisations reached matched fit (2.52–2.96 × $10^{-7}$; one was excluded at 3.4× the best fit). Their mean pairwise cosine was 0.927, minimum 0.835, and mean cosine to the truth 0.914. The effect replicated across four ground truths (Table II), with the same worst-disagreeing pair each time; recovered norm ratios clustered by initialisation to within about 0.01, so distinct solutions are reached systematically, not by chance.

**Table II.** Agreement among fit-matched solutions, four ground truths, 14 000 steps.

| Ground truth | Best fit (MSE) | Mean pairwise cosine | Min pairwise cosine |
|---|---|---|---|
| 0 | $3.51 \times 10^{-7}$ | 0.893 | 0.807 |
| 1 | $3.65 \times 10^{-7}$ | 0.893 | 0.815 |
| 2 | $3.11 \times 10^{-7}$ | 0.901 | 0.826 |
| 3 | $3.13 \times 10^{-7}$ | 0.900 | 0.819 |
| Mean ± SD | | 0.897 ± 0.004 | 0.817 ± 0.007 |

## C. Barriers, not a null space

With the five blind, fit-matched solutions of ground truth 0 (errors 3.57–3.90 × $10^{-7}$ after 14 000 steps; agreement mean 0.954, minimum 0.890), the error along the straight line rose between every one of the ten pairs, to 1.5–6.0× the worse endpoint (median 3.8×), and the rise grew as the pair became less similar (Table III). Repeated for four ground truths (40 pairs), it rose for every pair of distinct solutions (29 pairs with agreement below 0.99: 2.4–8.2×, median 4.8×), while the 11 near-identical pairs share one minimum (1.0–2.2×). The near-truth controls reached a 4.2 ± 0.3× lower error and higher recovery

(0.946–0.977 against 0.891–0.926), and the line from the best blind solution to the near-truth one rose 8.1 ± 3.2× (4.7–12.4×) above the blind end. The blind solutions are therefore distinct local minima, separated by barriers from one another and from a deeper minimum: the inverse is non-convex and the null-space account is rejected. An earlier version of this test selected its pair among all eight starts, which paired an under-converged solution; it is withdrawn, and with a fit-matched pair the change along d stays below random directions at every scale, as it must when both ends fit, so it cannot detect a barrier. With a real 16-s stimulus representation as ground truth (held-out story, three blind starts and a near-truth control, 3 000 steps), blind recovery was 0.43–0.60, blind solutions agreed only at 0.19–0.56, the straight paths between them rose 11–21×, and the near-truth control reached a 2.9× lower error 54× barrier away: the same picture, more pronounced.

**Table III.** Straight-path barrier between fit-matched blind solutions (ground truth 0 per pair, four-truth summary in the last three rows; 14 000 steps).

| Solution pair | Agreement | Peak error on path | Peak / worse end |
|---|---|---|---|
| Gaussian 1.0 – zeros | 0.927 | $2.36 \times 10^{-6}$ | 6.0× |
| Gaussian 1.0 – uniform | 0.890 | $1.79 \times 10^{-6}$ | 4.6× |
| Gaussian 0.01 – Gaussian 1.0 | 0.927 | $1.71 \times 10^{-6}$ | 4.4× |
| uniform – zeros | 0.958 | $1.50 \times 10^{-6}$ | 4.1× |
| Gaussian 0.1 – Gaussian 1.0 | 0.940 | $1.57 \times 10^{-6}$ | 4.0× |
| Gaussian 0.01 – uniform | 0.957 | $1.32 \times 10^{-6}$ | 3.6× |
| Gaussian 0.1 – uniform | 0.956 | $1.25 \times 10^{-6}$ | 3.4× |
| Gaussian 0.01 – Gaussian 0.1 | 0.992 | $8.12 \times 10^{-7}$ | 2.2× |
| Gaussian 0.1 – zeros | 0.993 | $5.52 \times 10^{-7}$ | 1.6× |
| Gaussian 0.01 – zeros | 0.999 | $5.57 \times 10^{-7}$ | 1.5× |
| Best blind (zeros) – near-truth | — | $2.58 \times 10^{-6}$ | 7.2× |
| Four truths: 29 distinct pairs | 0.89–0.99 | — | median 4.8× (2.4–8.2×) |
| Four truths: 11 near-identical pairs | ≥ 0.99 | — | 1.0–2.2× |
| Four truths: best blind – near-truth | — | — | 8.1 ± 3.2× |

### D. Error lies in the least-sensitive directions

On the least similar fit-matched pair (Gaussian 1.0 and uniform), sensitivity along their difference d was $8.10 \times 10^{-2}$, against a median of $1.68 \times 10^{-1}$ over 200 random directions (range $1.22–2.39 \times 10^{-1}$): 2.1× below the median and below every random direction sampled; along the actual error $\hat{x} - x$ it was 1.8× below; over four ground truths the two ratios were 2.15 ± 0.17× and 1.68 ± 0.17×. Nine of the ten blind

pairs lay below the median (1.1–2.1×); the exception is the nearly identical pair from the zero and 0.01 starts (agreement 0.999), whose difference is too small to probe reliably. The operator responds about half as strongly to the components that inversion fails to recover: the inverse is anisotropically ill-conditioned.

## E. Conditioning in numbers

**Table IV.** Further characterisation (whole cortex unless stated).

| Test | Result |
|---|---|
| Gain (rescale x by α) | Only α = 1 within the residual floor; a 10% rescale gives 7.5× the floor, so magnitude is identifiable |
| Measurement noise, 0 → 100% of BOLD SD | Recovery 0.836 → 0.298, monotone; retrieval stays 0.99 |
| Observed cortex, 50 → 20 484 vertices | Recovery 0.100 → 0.836; 2 000 random vertices (10%) reach 0.777 |
| $\ell_2$ prior (500 vertices) | Best +0.032 at λ = $10^{-5}$, harmful above |
| Temporal-smoothness prior (500 vertices) | Recovery 0.527 → 0.0008; retrieval to chance |
| Ridge decoder (240 train, 60 test) | Best test cosine 0.060, against 0.918 for gradient inversion |
| Stimulus manifold | White noise 0.947 ± 0.004 vs smooth 0.942 ± 0.005 |
| Rate of semantic change, 64-fold range | 0.836–0.858, no trend |

The ridge decoder predicts a time-averaged representation, a strictly easier target, so the gap is a lower bound.

## F. Predictable is not recoverable, region by region

Forward sensitivity follows anatomy (Table V): language and default-mode networks are dominated by text-derived dimensions (z = +26.6 to +28.5), auditory cortex by audio (+34.7) and visual cortex by video (+40.7). Recoverability does not follow sensitivity. Observing only its own vertices, visual cortex, the most responsive network (1.60× the cortical mean), recovered 0.125 worse than size-matched random vertices; somatomotor, the least responsive (0.73×), recovered 0.102 better and exceeded the whole cortex at the same budget (0.851 vs 0.836). Both effects are about ten times the control spread (second random seed: 0.0014 and 0.0126). Response diversity explains the extremes: somatomotor alone is more diverse than its control (participation ratio 18.6 vs 15.2), while visual cortex is concentrated (10.3; top direction 27% of variance).

**Table V.** Network sensitivity and recoverability (synthetic, 8 000 steps).

| Network | Vertices | Sensitivity (× cortex) | Dominant slot (z) | Recovery, network | Recovery, random | Difference | Participation ratio (network / control) |
|---|---|---|---|---|---|---|---|
| Somatomotor | 1 487 | 0.73 | none | 0.851 | 0.748 | +0.102 | 18.6 / 15.2 |
| Language (L) | 1 282 | 1.21 | text +26.6 | 0.752 | 0.729 | +0.023 | 9.7 / 15.0 |
| Auditory | 277 | 1.29 | audio +34.7 | 0.386 | 0.371 | +0.015 | 10.1 / 15.5 |
| Language (R) | 1 368 | 1.22 | text +28.5 | 0.750 | 0.739 | +0.011 | 13.5 / 15.1 |
| Default mode | 1 224 | 1.03 | text +26.9 | 0.727 | 0.723 | +0.004 | 11.9 / 15.0 |
| Visual | 1 582 | 1.60 | video +40.7 | 0.629 | 0.754 | −0.125 | 10.3 / 15.1 |

### G. Real fMRI

The alignment offset fitted on the training stories was 4 s with text + audio features (6 s with audio alone). Table VI reports the held-out story; the text + audio run (Colab T4, 2 000 inversion steps) is primary, and the earlier audio-only run (CPU, 600 steps) is shown for comparison.

**Table VI.** Real-fMRI validation, held-out story *wheretheressmoke* (UTS03).

| Measure | Text + audio | Audio only | Chance / ceiling |
|---|---|---|---|
| Encoding r, whole cortex | 0.199 | 0.062 | 0 |
| Encoding r, top 2 000 vertices | 0.478 (56% of ceiling) | 0.273 (32%) | ceiling 0.859 |
| Encoding r, auditory / language L / language R | 0.405 / 0.274 / 0.348 | 0.263 / 0.193 / 0.200 | 0 |
| Encoding r, default mode / somatomotor / visual | 0.348 / 0.164 / 0.176 | 0.103 / 0.039 / −0.020 | 0 |
| Forward identification, 20-s windows, top-1 (top-5) | 58.6% (86.2%) | 20.7% (41.4%) | 0.5% (194 windows, 29 queries) |

| Measure | Text + audio | Audio only | Chance / ceiling |
|---|---|---|---|
| Inversion of measured BOLD, 100-s segments, top-1 | 4 / 5 (ranks 0, 0, 6, 0, 0) | 5 / 5 | 2.2% (45 segments) |
| Cosine of $\hat{x}$ to the true x | 0.024 | 0.062 | — |
| Oracle: inverting the model's own prediction, top-1 / cosine | 5 / 5 / 0.224 | — | — |

Adding language features more than doubled encoding accuracy on the held-out story and nearly tripled forward identification, with language and default-mode networks gaining most. Inverting measured BOLD ranked the true segment first in 4 of 5 held-out queries (the fifth ranked seventh of 45), including against the six other segments of the same story, while $\hat{x}$ kept a cosine of only 0.02 to the truth. The oracle separates the two limits. Without noise, inversion identifies every segment yet still recovers only 0.22 of the representation, so most of the loss is the operator's conditioning, not measurement noise; noise then lowers recovery further while leaving identification largely intact, as in the synthetic noise sweep (Table IV). On real data, as in simulation, the inverse identifies which moment was heard but does not reconstruct its representation.

## V. Discussion

**Identification is robust; reconstruction is not.** The synthetic and real results tell one story. Under noise, retrieval stays near 0.99 while cosine falls to 0.30; on real fMRI, inversion identifies four of five held-out segments while recovering a representation with cosine 0.02, and even noise-free inversion of the model's own prediction recovers only 0.22. Identification needs only the high-sensitivity components that separate one moment from another, and those are the components the operator transmits well. The components it transmits weakly are where recovery fails (Section IV-D). Decoding claims built on identification accuracy should therefore not be read as reconstruction of the underlying representation.

**Why better encoders need not decode better.** Encoding accuracy is a property of f near the data; decodability depends on how f treats directions in input space. A better fit can come almost entirely from the high-sensitivity directions, leaving the low-sensitivity ones as ill-determined as before. This is the encoding–decoding dissociation in Section IV-A, and it means scaling laws for encoding [3] cannot be assumed to carry over to decoding.

**The limit is geometry, not lost information.** Fit-matched solutions are separate minima, not points on a flat valley, and magnitude is identifiable. Recovery is limited by a non-convex, anisotropically conditioned landscape, which better optimisation, initialisation or priors matched to the operator's weak directions could improve. Generic priors do not: $\ell_2$ helps marginally and temporal smoothness destroys recovery.

**Regions.** The most responsive network is the worst to decode from because its response is concentrated in few directions; the least responsive is the best because its response is diverse. A uniform whole-cortex loss is therefore dominated by high-gain, low-diversity regions, and down-weighting them is a testable way to improve decoding.

**Relation to linear theory.** For linear inverted encoding models, the stimulus is recovered only up to an invertible transform [9]. A deep encoder removes that closed-form ambiguity but replaces it with a non-convex, direction-dependent one that has to be measured empirically, as done here.

### Limitations

1. Synthetic ground truth lives in a random low-dimensional subspace, not in real language embeddings; qualitative findings transfer to real data (Section IV-G), absolute values may not.
2. Inversion targets the aggregated representation, one stage after the raw extractor features.
3. All solutions use Adam; SGD and L-BFGS did not reach comparable fit under the settings tried. The non-convexity claim rests on optimiser-independent forward-pass tests and on replication.
4. The released model has no explicit haemodynamic operator and ships only the average-subject layer, so temporal bandwidth and per-subject analyses are out of reach with public weights.
5. Regional z-scores use a random-vertex null that ignores spatial autocorrelation; a spin test is stricter. Regional inversion uses one ground truth and one budget.
6. The real-data test covers one subject and five held-out inversion queries; regional recoverability is established in simulation, and a per-network test on real data needs held-out stories from more subjects, since the other five stories were in TRIBE v2's training set; 100-s inversion segments and 20-s forward windows are not directly comparable. One training story (*avatar*) showed no alignment peak and was kept only as a distractor. Public fsaverage5 meshes stand in for TRIBE v2's internal surfaces.

### Conclusion

A near-perfect deep brain encoder does not by itself make meaning recoverable from fMRI. Inverting TRIBE v2 is a non-convex, direction-dependently conditioned problem: it identifies which stimulus was presented, on synthetic and real data alike, but reconstructs only the components the operator transmits well. Recoverability has to be measured, not inferred from predictivity, and it differs across cortex in ways predictivity does not reveal.

## Data and Code Availability

TRIBE v2 weights and code are public (Meta FAIR, CC BY-NC 4.0). fMRI data: OpenNeuro ds003020 [14]. All analysis code, the fsaverage5 sampling matrix, result files and the Colab notebook that reproduces the real-data analysis: github.com/Ivxn404/semantic-decoding-tribe-inversion.

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## References

1. T. Naselaris, K. N. Kay, S. Nishimoto, and J. L. Gallant, "Encoding and decoding in fMRI," *NeuroImage*, vol. 56, no. 2, pp. 400–410, 2011.

2. A. G. Huth, W. A. de Heer, T. L. Griffiths, F. E. Theunissen, and J. L. Gallant, "Natural speech reveals the semantic maps that tile human cerebral cortex," *Nature*, vol. 532, pp. 453–458, 2016.
3. R. Antonello, A. Vaidya, and A. G. Huth, "Scaling laws for language encoding models in fMRI," in *Proc. NeurIPS*, vol. 36, 2023.
4. J. Tang, A. LeBel, S. Jain, and A. G. Huth, "Semantic reconstruction of continuous language from non-invasive brain recordings," *Nat. Neurosci.*, vol. 26, pp. 858–866, 2023.
5. S. d'Ascoli *et al.*, "A foundation model of vision, audition, and language for in-silico neuroscience," arXiv:2605.04326, 2026.
6. S. d'Ascoli, J. Rapin, Y. Benchetrit, H. Banville, and J.-R. King, "TRIBE: TRImodal Brain Encoder for whole-brain fMRI response prediction," in *Proc. ICLR*, 2026.
7. K. N. Kay, T. Naselaris, R. J. Prenger, and J. L. Gallant, "Identifying natural images from human brain activity," *Nature*, vol. 452, pp. 352–355, 2008.
8. T. M. Mitchell *et al.*, "Predicting human brain activity associated with the meanings of nouns," *Science*, vol. 320, pp. 1191–1195, 2008.
9. J. L. Gardner and T. Liu, "Inverted encoding models reconstruct an arbitrary model response, not the stimulus," *eNeuro*, vol. 6, no. 2, ENEURO.0363-18.2019, 2019.
10. Bladon and Bent, "Feature visualization recovers known cortical selectivity from TRIBE v2," arXiv:2605.13904, 2026.
11. Y. Benchetrit *et al.*, "Boosting brain-to-image decoding with TRIBE v2 data augmentation," arXiv:2606.06345, 2026.
12. P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*. Philadelphia, PA, USA: SIAM, 2010.
13. J. Hadamard, "Sur les problèmes aux dérivées partielles et leur signification physique," *Princeton Univ. Bull.*, vol. 13, pp. 49–52, 1902.
14. A. LeBel *et al.*, "A natural language fMRI dataset for voxelwise encoding models," *Sci. Data*, vol. 10, art. 555, 2023.
15. A. Défossez, C. Caucheteux, J. Rapin, O. Kabeli, and J.-R. King, "Decoding speech perception from non-invasive brain recordings," *Nat. Mach. Intell.*, vol. 5, pp. 1097–1107, 2023.
16. J. Lévy, M. Zhang, S. Pinet, J. Rapin, H. Banville, S. d'Ascoli, and J.-R. King, "Brain-to-text decoding: A non-invasive approach via typing," arXiv:2502.17480, 2025.
17. P. S. Scotti *et al.*, "MindEye2: Shared-subject models enable fMRI-to-image with 1 hour of data," in *Proc. ICML*, 2024.
18. D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2015.
19. C. Destrieux, B. Fischl, A. Dale, and E. Halgren, "Automatic parcellation of human cortical gyri and sulci using standard anatomical nomenclature," *NeuroImage*, vol. 53, no. 1, pp. 1–15, 2010.
20. E. Fedorenko, A. A. Ivanova, and T. I. Regev, "The language network as a natural kind within the broader landscape of the human brain," *Nat. Rev. Neurosci.*, vol. 25, pp. 289–312, 2024.
21. R. L. Buckner, J. R. Andrews-Hanna, and D. L. Schacter, "The brain's default network," *Ann. N.Y. Acad. Sci.*, vol. 1124, pp. 1–38, 2008.
22. G. J. Brouwer and D. J. Heeger, "Decoding and reconstructing color from responses in human visual cortex," *J. Neurosci.*, vol. 29, no. 44, pp. 13992–14003, 2009.
23. P. Gao *et al.*, "A theory of multineuronal dimensionality, dynamics and measurement," bioRxiv 214262, 2017.

24. B. Fischl, M. I. Sereno, R. B. H. Tootell, and A. M. Dale, "High-resolution intersubject averaging and a coordinate system for the cortical surface," *Hum. Brain Mapp.*, vol. 8, no. 4, pp. 272–284, 1999.
25. B. B. Avants, C. L. Epstein, M. Grossman, and J. C. Gee, "Symmetric diffeomorphic image registration with cross-correlation," *Med. Image Anal.*, vol. 12, no. 1, pp. 26–41, 2008.
26. J. S. Gao, A. G. Huth, M. D. Lescroart, and J. L. Gallant, "Pycortex: an interactive surface visualizer for fMRI," *Front. Neuroinform.*, vol. 9, art. 23, 2015.
27. Y.-A. Chung *et al.*, "W2v-BERT: Combining contrastive learning and masked language modeling for self-supervised speech pre-training," in *Proc. IEEE ASRU*, 2021, pp. 244–250.
28. A. Grattafiori *et al.*, "The Llama 3 herd of models," arXiv:2407.21783, 2024.
29. A. Mahendran and A. Vedaldi, "Understanding deep image representations by inverting them," in *Proc. IEEE CVPR*, 2015, pp. 5188–5196.
30. A. T. Gifford et al., "The Algonauts Project 2025 challenge: How the human brain makes sense of multimodal movies," arXiv:2501.00504, 2025.
31. B. Lahner et al., "Modeling short visual events through the BOLD Moments video fMRI dataset and metadata," Nat. Commun., vol. 15, art. 6241, 2024.
32. H. Wen, J. Shi, Y. Zhang, K.-H. Lu, J. Cao, and Z. Liu, "Neural encoding and decoding with deep learning for dynamic natural vision," Cereb. Cortex, vol. 28, no. 12, pp. 4136–4160, 2018.
33. N. K. Logothetis, "What we can do and what we cannot do with fMRI," Nature, vol. 453, pp. 869–878, 2008.
34. C. M. Michel, M. M. Murray, G. Lantz, S. Gonzalez, L. Spinelli, and R. Grave de Peralta, "EEG source imaging," Clin. Neurophysiol., vol. 115, no. 10, pp. 2195–2222, 2004.
35. Q. K. Telesford et al., "An open-access dataset of naturalistic viewing using simultaneous EEG-fMRI," Sci. Data, vol. 10, art. 554, 2023.